

Marcelo Guzman

MONITORING & GEOSPATIAL SPECIALIST

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EXPERIENCE

Landhawk Consulting LLC

GIS & Reporting Specialist

Sep 2026 – Present

Provides GIS development, spatial data management, technical reporting, and QA/QC involving natural resources, cultural resources, water availability, and environmental compliance for infrastructure projects across Texas, New Mexico, and Arizona.

- Developed and deployed a Python application to automate photolog metadata extraction, eliminating ~12 hours of manual data entry per technician per week and generating an estimated \$5,000+ in weekly labor savings.

IRCA

Environmental Compliance Coordinator

Sep 2025 – Sep 2026

Apr 2026 – Sep 2026

Lead compliance monitoring and geospatial verification programs, supervise field teams across a portfolio of 12+ industrial clients in construction, steel, forging, foundry, ceramics, and automotive & aerospace components.

- Expanded the firm's service portfolio with four new revenue lines, generating USD \$85,000+ in new annual service revenue: UAV-based environmental monitoring reports; photogrammetry and remote sensing (DEM/DTM/DSM, orthoimagery, contour and slope mapping, supervised classification); reforestation assessments; and biological survey design and execution; plus, recurring monthly supervision retainers.
- Shielded 4 clients from fines and operational shutdown by authoring their Environmental Legal Requirements Matrices, mapping every applicable federal, state, and municipal obligation to plant operations, then assembling the required licenses and permits into dossiers for inspection defending.
- Produced annual ESG due diligence & performance reports for major NA steel producers, benchmarking operations against the IFC Performance Standards and delivering the E&S baselines, mitigation plans, and management systems required to satisfy international investor requirements.
- Directed GIS and UAV remote-sensing operations across 56+ drone missions (~556 acres surveyed), verifying 92% reforestation survival against an 80% permit threshold and mapping >45% steep-slope no-build zones on 5 construction sites.
- Cleared a backlog of 16 compliance reports unfiled for ~9 years, reducing client penalties from suspension to minor warnings.

Project Manager

Sep 2025 – Apr 2026

Managed ~\$210K USD in contracts delivering state and federal environmental permitting and audits for clients in steel, iron & aluminum foundries, machining, automotive and aerospace parts, and construction analyzing operational impacts.

- Delivered 30+ federal & state regulatory instruments in 10 months at a 90%+ on-time rate against target deadlines, including: 9 environmental impact statements, 7 operating licenses, 3 wastewater-discharge registrations, 3 land-use-change technical studies, and 4 air-emissions duct & stack studies.
- Launched the first photogrammetry line, eliminating the need to outsource surveys and saving ~US\$6,000 per compensation project.
- Ran weekly regulatory audits for ~22 construction and manufacturing clients, leading inspection responses and corrective actions.
- Built and published two R/Shiny applications that standardize recurring impact assessment workflows for consulting teams.

Enel Green Power

Ecological Data Analyst

Jun 2024 – Feb 2025

Conducted pre- and post-construction ecological surveys and geospatial data pipelines for wind farm projects.

- Automated the transformation of raw camera-trap metadata into analysis-ready species databases in R, cutting a process that took technicians ~48 hours to under 30 minutes supporting monitoring of sensitive species (lynx, Texas tortoise) and large mammals across wind farms.

EDUCATION

Graduate certificate Geoinformatics, University of Debrecen · GPA 4.0/4.0 (Stipendium Hungaricum Scholar)

BS Biology, Universidad Autónoma de Nuevo León · GPA 3.8/4.0 (Honors Program)

International Baccalaureate (IB) Diploma

PROJECTS

- **Biodiversity Baseline app** · published R/Shiny, *IRCA*: automates biodiversity baselines for EIA exporting lists and APA cited references for GIS.
Skills: R, R/Shiny, rgbif, CSS, HTML, JS, GitHub
- **Toxic Waste Analytics app** · published R/Shiny, *IRCA*: summarizes Pollutant Release and Transfer Register data for client process reviews.
Skills: R, R/Shiny, CSS, HTML, JS, GitHub
- **Risk Modeling** · published GitHub Pages, *DATATON*: city fire risk analysis and susceptibility modeling with commercial satellite imagery.
Skills: Google Earth Engine, R, Python, CSS, HTML, JS, ArcGIS Online, StoryMaps, GitHub
- **Urbanization Index**, *EU research internship*: geospatial workflow that grades urban to rural gradients for ornithology research
Skills: R, Python, ArcGIS Pro, ArcGIS ModelBuilder, QGIS, SQL, Excel
- **UAV Accuracy Assessment Projects**, *University of Debrecen*: benchmarked accuracy across GCP and RTK processing pipelines
Skills: Emlid RTK, Network RTK, ENVI, Pix4D Enterprise, ArcGIS Pro, photogrammetry, DEM/DSM